the Journal of Strength and Conditioning Research™

Considerations for Velocity-Based Training: The Instruction to Move "As Fast As Possible" Is Less Effective Than a Target Velocity

Steven M. Hirsch and David M. Frost

Faculty of Kinesiology and Physical Education, University of Toronto, Toronto, Ontario, Canada

Abstract

Hirsch, SM and Frost, DM. Considerations for velocity-based training: the instruction to move "as fast as possible" is less effective than a target velocity. *J Strength Cond Res* XX(X): 000–000, 2019—In addition to the potential benefits with monitoring training intensity with velocity-measuring tools during exercise, these devices also provide an opportunity for researchers and practitioners to provide feedback and instruction to performers in a new way. The purpose of this study was to assess the influence of instructing athletes with a target velocity vs. instructing them to move "as fast as possible" during a free-weight bench press. In addition, the effects of each instruction on future repetition maximum (RM) test performance was compared. Thirteen male powerlifters were recruited and performed 4 sets of 5 repetitions with 45% 1RM while being instructed to attain a target velocity of 1.0 $m \cdot s^{-1}$ or to move as fast as possible. The maximum mean velocity attained in each set was compared with a repeated measures analysis of variance and Tukey's post-hoc test. After these 4 sets of 5 repetitions, the number of repetitions performed during an RM test with 75% 1RM was compared with a Wilcoxon Signed-Rank test. Participants moved faster when they received the target velocity instruction (0.84 ± 0.10 $m \cdot s^{-1}$) than when they received the "move as fast as possible" instruction (0.82 ± 0.09 $m \cdot s^{-1}$) ($p < 0.001$) with no differences in the number of repetitions performed in the following RM test between the 2 testing sessions ($p = 0.43$). An instruction to attain a challenging velocity target may be a more effective strategy to use when attempting to elicit maximum barbell velocities during training relative to the traditional instruction to move as fast as possible.

Key Words: speed, power, feedback, bench press, exercise

Introduction

With the increased popularity of various commercial linear position transducers and inertial measurement units, collecting biomechanical data to assist with making training decisions has become easier for researchers and exercise professionals. For example, barbell velocity data can be quantified with these devices and provide an indication of a performer's state of readiness. Several researchers have reported a strong association between the relative load used and the absolute velocity attained, for a given relative load, making it possible to estimate a performer's maximum strength (6,9,17,19). Decreases in intra-set squat and bench press velocity have been highly associated with lactate and ammonia concentrations ($r = 0.95$–97, $p < 0.001$) (18), suggesting that velocity drop-off during a set can provide an indication of the accumulation of metabolic by-products associated with increased fatigue. Furthermore, given that the velocity attained during a 1 repetition maximum (1RM) attempt is not statistically different from the velocity attained during the last repetition of an RM test (xRM) for a given exercise (8), velocity data may help estimate volitional failure during a training session.

In addition to the potential benefits provided by monitoring velocity while training, how athletes are instructed and provided with feedback can also influence the effectiveness of an exercise intervention. An example of augmented feedback that can be provided to the performer is knowledge of results (KRs), which provides information regarding the outcome of their movement (10). Knowledge of result can be further divided into qualitative (i.e., feedback about the quality of movement) and quantitative KR (i.e., numerical feedback about the outcome of movement) (20) to suit the intended training objective. While providing quantitative KR, it is also possible to establish specific goals for an athlete to achieve within a training session. It has been suggested that specific goals can provide motivation for people to attain higher levels of performance (12), especially when they are challenging (11). These specific, challenging goals may motivate people more effectively than an instruction to just "do their best", (11) as this is defined idiosyncratically and does not provide any external or objective reference.

While linear position transducers and inertial measurement units make it possible for training interventions to become data-driven, they also provide an opportunity for researchers and practitioners to provide feedback and instruction to athletes in a new way. As a combination of both quantitative and qualitative KR could result in superior skill acquisition than only using a single method (10), instructions that use velocity data (quantitative KR) may enhance the velocity attained during training. Providing feedback of how fast performers were able to move during a squat jump has been shown to improve vertical and horizontal jump performance, reduce sprint times (14), and decrease the variability in velocity attained between repetitions during training (15). This may result in athletes attaining higher movement velocities more often during training, which could bring about superior training adaptations. Randell et al. (15) also reported that athletes may attain faster movement velocities

Address correspondence to Dr. David M. Frost, d.frost@utoronto.ca.

Journal of Strength and Conditioning Research 00(00)/1–6

during training if provided feedback of their movement velocity. However, it is not known whether the movement velocity attained during a free-weight bench press would be greater if providing athletes with a target velocity (i.e., specific goal) to attain. It is also unknown whether the need to supply greater energy while attaining faster movement velocities may cause an athlete to perform fewer repetitions during future sets of a training session.

Therefore, the purpose of this study was to compare the influence of 2 instructions on bench press velocity (i.e., move as fast as possible vs. achieve a target velocity of 1.0 $m\cdot s^{-1}$). The effects of each instruction on performance in an RM test were compared to examine the influence of each during future sets of a training session. It is hypothesized that participants will attain a higher mean velocity when being instructed to attain the target velocity, and would perform fewer repetitions during an RM test after this instruction protocol.

Methods

Experimental Approach to the Problem

Thirteen male powerlifters were recruited for this investigation. Participants completed 2 testing sessions. In session A, participants were cued to attain a mean velocity of 1.0 $m\cdot s^{-1}$ while performing a free-weight bench press with 45% 1RM. In session B, participants were cued to lift a 45% load as fast as possible. This relative load was chosen as it was the load where participants produced the largest mean power in the investigation by Frost et al. (5) The order of each session was randomized. Participants were required to avoid any 1RM or RM tests one week before their first scheduled testing session. The second session was performed 3–7 days after the completion of the first testing session. Each session involved a general and specific warm-up, a maximum velocity test, a 1RM test, 4 sub-maximal "velocity" sets, and an RM test. The maximum velocity test consisted of 3 sets of one repetition with a wooden dowel. Participants were given a maximum of 5 attempts to establish their maximum bench press strength. After the 1RM test, participants performed 4 sets of 5 repetitions with 45% 1RM and were provided with the instruction to either attain a target velocity of 1.0 $m\cdot s^{-1}$ (session A) or to move as fast as possible (session B). The mean velocity of all repetitions was recorded with a 3-dimensional motion capture system. In both testing sessions, participants were given feedback regarding their barbell velocity after each repetition. Feedback was provided on a tablet via a GymAware linear position transducer.

Subjects

Fifteen male powerlifters completed testing; however, only 13 were included in the study (mean ± *SD*: age: 26.8 ± 7.8 years, height: 175.0 ± 6.9 cm, mass: 87.2 ± 14.3 kg). One excluded powerlifter did not perform both testing sessions within the 3–7-day period, whereas another self-reported excessive fatigue from the previous session and displayed a 1RM difference of 10 kg. The inclusion criteria for the participants was the completion of at least one sanctioned powerlifting competition. The average training experience of the 13 participants was 7.5 ± 3.9 years and the average years of competition experience was 2.0 ± 0.9 years. This study was approved by the ethics review board of the University of Toronto. Participants signed an informed consent form and a PAR-Q and YOU form before the commencement of testing.

Procedures

Participants were given 5 minutes to perform their own preferred low intensity warm-up and/or flexibility routine before the commencement of any bench press exertions. This general warm-up protocol was recorded, so that it could be repeated before both testing sessions. After the general warm-up, a series of standardized instructions were provided: (a) the head and hips must remain in contact with the bench and the feet must be kept flat on the floor throughout each repetition; (b) the barbell must be lowered within 0.05 m of the chest, but must not contact the chest; and (c) each repetition must be performed without pausing between the lifting and lower phases. Participants' grip and foot width were self-determined and marked before any repetitions with the barbell for future reference. Both measurements were kept consistent throughout the 2 testing sessions. If any repetition was disregarded, participants received feedback as to why it did not meet the study criteria. Participants were given 5minutes of rest to repeat their attempt during the 1RM test under the given constraints. However, participants were not given an opportunity to repeat any sub-maximal exertions.

After receiving the instructions, a standardized warm-up with the barbell was performed. Participants performed 5 repetitions with an unloaded barbell (20 kg), 35 and 50% of an estimated 1RM. Two minutes of rest was provided between each set. Participants then performed a maximum velocity test with a wooden dowel (approximately 0.26 kg) to estimate their maximum velocity (i.e., 0% 1RM). Participants performed 3 repetitions as fast as possible with the dowel, each separated by 30 seconds of rest. Then, using a protocol similar to Frost et al. (5), participants progressed to a 1RM bench press. Participants performed one set of 4 repetitions with 60% of an estimated 1RM, one set of 3 repetitions with 70% of an estimated 1RM, one set of 2 repetitions with 80% of an estimated 1RM, and one set of one repetition with 90% of an estimated 1RM. Participants were then given a maximum of 5 attempts to determine their 1RM. Three minutes of rest was provided between each set with loads less than 90% of the estimated 1RM, and 5 minutes of rest was provided for all sets exceeding 90% of the estimated 1RM.

After 5 minutes of rest, participants began their sub-maximal "velocity" sets. In both testing sessions, participants performed 4 sets of 5 single repetitions with 45% of their measured 1RM. Fifteen seconds and three minutes of rest were provided between each repetition and set, respectively. During the testing of session A participants were instructed to lift at a mean velocity of 1.0 $m\cdot s^{-1}$ during the lifting phase of the bench press. This velocity was chosen, as it represented a challenging velocity target for 45% 1RM based on load–velocity regression equations from Frost et al. (5) and Helms et al. (7). After each repetition, participants were told the barbell velocity. Feedback of barbell velocity was provided with a GymAware (GymAware Power Tool; Kinetic Performance Technologies, Canberra, Australia) linear position transducer attached directly beneath the right collar of the barbell. During the testing of session B, participants were instructed to complete the lifting phase as fast as possible (AFAP). Similar to testing session A, participants were told their barbell velocity after each repetition. The order of each session (instruction) was randomized between participants. Following 5 minutes of rest, participants performed as many repetitions as possible with 75% of their measured 1RM. No instructions or feedback pertaining to the lifting velocity was provided.

The mean velocity of each repetition was measured with a three-dimensional motion capture system (Optritrack Flex 13;

NaturalPoint Inc., Corvallis, OR, USA). The ADTech Motion Analysis Software System (AMASS; C-Motion, Germantown, MD) was used to derive the 3-dimensional locations of a spherical retroreflective marker placed on the center of the barbell. Three cameras were placed around the barbell to capture the position of this marker at a sampling frequency of 120 Hz. Marker displacement data were smoothed with a fourth order, dual pass Butterworth filter in Visual3D (C-Motion) with a cut-off frequency of 10 Hz. Event detection algorithms were used to define the bottom and top position of each repetition to isolate the lifting (i.e., concentric) phase. The bottom position of each repetition was defined as the lowest point of displacement, and the top position of each repetition was defined as the global maximum point of displacement following the lowest point. The mean lifting velocity was computed by dividing the vertical displacement by the time between the maximum and minimum events.

Statistical Analyses

The fastest mean velocity attained with a 45% 1RM load during each 5-repetition set was the dependent variable for the first question in this investigation. The fastest mean velocity for each 5-repetition set was compared with a 2 × 4 (instruction × set number) repeated-measures analysis of variance. Mauchlys test of sphericity was used to ensure equality of variance across all levels of the within-subject factors. Tukey's post-hoc test was used to identify any significant differences between factors. Statistical significance was set at $p \leq 0.05$. Effect sizes (ω^2) were also computed for significant main effects.

The number of repetitions performed with the 75% 1RM load was the dependent variable used to answer the second question. The instruction used before the RM test constituted the within-participant factor. To assess the effects of the instruction on a future RM test, a Wilcoxon Signed-Rank Test was used to compare the number of repetitions performed in each testing session. This statistical test was used in lieu of a paired-samples *t*-test as a Shapiro–Wilk test showed that the data were not normally distributed (p = 0.004). Statistical significance was set at $p \leq 0.05$. All statistical analyses were performed with the R programming language.

Results

The average 1RM for participants during session A and B was 140.0 ± 23.6 kg and 139.0 ± 23.3 kg, respectively. Instructing participants to achieve a target velocity prompted a significantly higher mean velocity (0.84 ± 0.10 $m \cdot s^{-1}$) in comparison to encouraging them to move as fast as possible (0.82 ± 0.09 $m \cdot s^{-1}$) ($F = 14.91$, df = 1, 84, $p = 0.0002$, $\omega^2 = 0.29$). The fastest mean velocity achieved during sets 2, 3, and 4 were also found to be significantly higher than set 1, ($F = 10.42$, df = 3, 84, $p < 0.0001$, $\omega^2 = 0.60$; $p = < 0.001$–0.03); however, there was no significant interaction effect found ($F = 0.70$, df = 3, 84, $p = 0.58$). Participant-specific maximum mean velocities were found to be higher (+0.03 ± 0.02 $m \cdot s^{-1}$) when asked to achieve a target of 1.0 $m \cdot s^{-1}$ in 11 of the 13 participants (Figure 1). One participant exceeded this 1.00 $m \cdot s^{-1}$ target, with one more exceeding 0.95 $m \cdot s^{-1}$ and 4 more exceeding 0.90 $m \cdot s^{-1}$. There were no differences found in RM performance between the 2 testing sessions (Target Velocity Session = 9.61 ± 2.06 repetitions, AFAP Session = 10.0 ± 1.29 repetitions, V = 15.5, $p = 0.43$) (Figure 2).

Discussion

The purpose of this study was to assess the influence of 2 velocity-specific instructions during a free-weight bench press. This investigation also sought to evaluate whether subsequent performance in an RM test was impaired to the same extent with both instructions. The results showed that by providing performers with a challenging and specific target velocity, they were able to attain faster lifting velocities than when instructed to lift as fast as possible. Furthermore, despite reaching higher lifting velocities when instructed with a target, subsequent performance in an RM test was unaffected. This implies that a target may provide benefit acutely while limiting the negative impact on future sets within the training session.

The way in which athletes are instructed and provided with feedback can influence their kinematics and kinetics during exercise tasks. This has been noted in various other studies that have demonstrated that the instruction and feedback provided to athletes can improve their performance and acquisition of motor skills (10,21,23). For example, participants in other studies have been shown to jump higher (23) and produce a greater jump impulse and lower extremity joint moments (21) with an external focus of attention condition (focus on touching specific rings of a Vertec device) relative to an internal focus of attention (focusing on the finger to touch the rings of a Vertec) or a control condition (jumping as high as possible). A target velocity likely helps to facilitate an external focus of attention, which is important for improving performance and learning (22). In addition, by providing a challenging training goal, participants were provided context of how their actual movement velocity was, when compared with a standard while receiving KRs from the GymAware. The instruction to move as fast as possible is defined individually (11), and without any context participants may have only focused on beating or maintaining their previously attained velocity while receiving basic KR. However, by providing this external and specific reference with a target velocity, participants may have focused on trying to attain this target and were not limited by their individually defined goal of trying to beat or maintain their previous velocity. Furthermore, this specific velocity was chosen, as it would be difficult for powerlifters to achieve with 45% of their 1RM, and it has been reported that specific and challenging goals are important for improving performance (11,12). This study demonstrates that using a target velocity may increase movement velocity during training. Knowledge of result on its own may also provide some benefit, as after the first set, participants moved the barbell faster during subsequent sets, regardless of the instructions used.

Coker (3) conducted a similar study investigating horizontal jump distance when athletes were instructed with (a) no focus (jump as far as possible), (b) internal focus (rapid extension of knees), (c) external far (focus on jumping as close as possible to a cone placed 3 m away), and (d) external attainable (jumping as close as possible to a cone placed at their previous best distance). It was found that horizontal displacement was longer when instructed with the attainable target vs. all other conditions. Although participants did not jump as far with the “external far” instruction, it may have been that this goal was too challenging, and thus the commitment to the goal may have lapsed (4). This study highlights that it is important for the target provided to the athletes to be attainable. In this investigation, participants moved at 0.84 $m \cdot s^{-1}$ on average when they were given the target velocity instruction with only one participant actually exceeding the velocity target. Although the goal was not attained by most participants, given that they were still moving at 84% of the target velocity they may not have not lost motivation to try to attain that actual target, resulting in them moving faster. However, future work could look to use individualized velocity targets that may

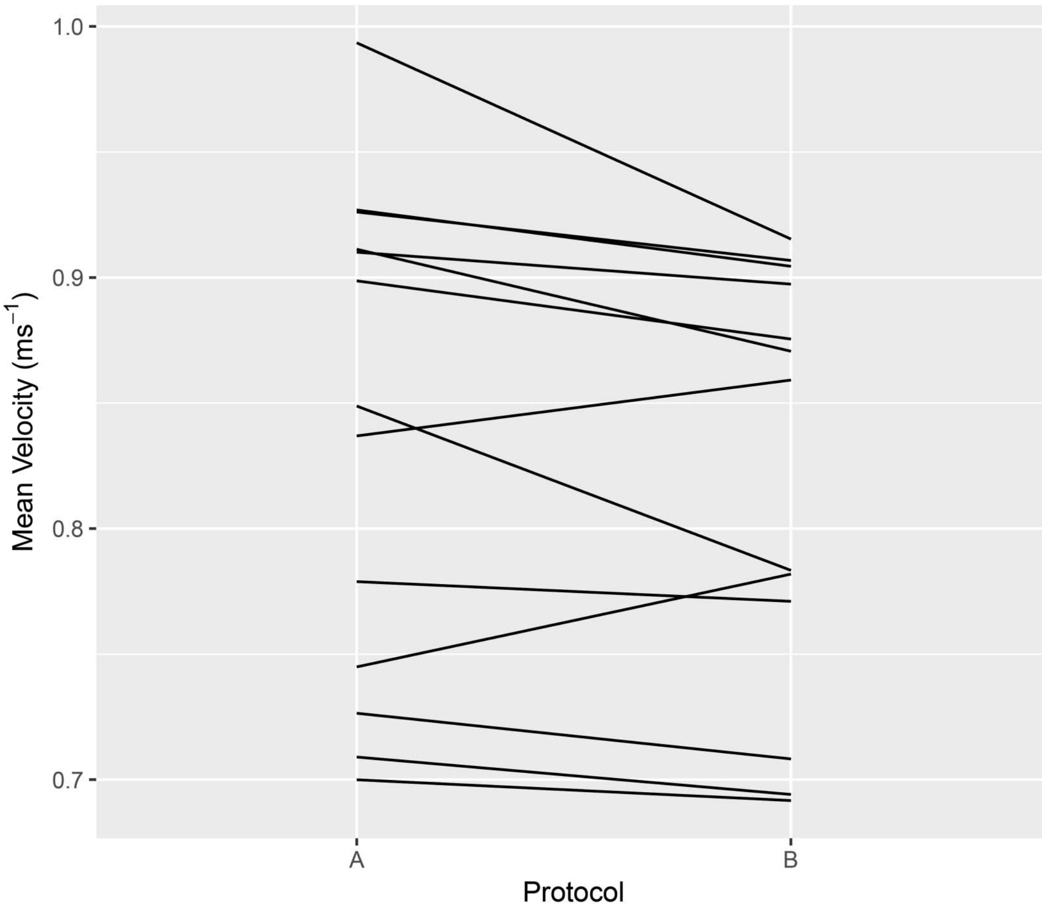

Figure 1. Participants' averaged mean velocity across the 4 sets with 45% 1 repetition maximum. Protocol A represents the instruction to attain a target velocity of 1.0 m·s^{-1}. Protocol B represents the instruction to move as fast as possible. Eleven of 13 participants moved faster (as an average of their 4 sets) when instructed to attain a target velocity of 1.0 m·s^{-1}.

better accommodate the abilities of each performer. This could potentially bring about even faster movement velocities during training.

Although the athletes increased their bench press velocity when receiving the target velocity instruction, there was no difference in subsequent RM performance when compared with the as fast as possible condition. It was hypothesized that by increasing movement velocity during each subsequent set, more energy would be needed, and thus there would be greater physiological fatigue. However, this was not seen. There may be several factors that influenced this result. For example, it is possible that while greater energy was expended when instructed to attain a target velocity, because the participants moved quicker, this may have brought about a neural potentiation effect (16). Therefore, any additional fatigue incurred from the increased energy expenditure may have been offset by the subsequent excitation of the nervous system from attaining faster movement velocities. It may have also been that participants were statistically faster when provided with the target velocity instruction vs. the "move as fast as possible" instruction, but not fast enough for increased muscle fatigue to occur. The average difference across all participants was an increase in 0.02 m·s^{-1}, which may have been too little for any practically meaningful accumulation of fatigue to occur. In addition, the method of assessing differences in movement velocities may have also resulted in statistically different movement velocities as only the maximum mean velocities of each set were analyzed. However, if all repetitions in a set were considered in the analysis (as with an average mean velocity of the set), there may have been no statistically significant differences. If this were true, then there may not have been an increased accumulation of fatigue if instructed to attain a target velocity.

Velocity feedback alone (quantitative KR) can reduce the between-rep variability in movement velocity during exercise, while also potentially increasing movement velocity (14); but providing a challenging target can also bring about superior changes in movement velocity. The increase in velocity in this investigation was found even though both groups received KR after each repetition. Specifically, 11 of 13 participants attained a higher max velocity when instructed to reach a target of 1.0 m·s^{-1} (Figure 1). These 11 participants moved, on average, 0.03 m·s^{-1} faster when given the target velocity instruction. Attaining these faster movement velocities is important if trying to prioritize fast movement velocities during training to bring about superior high velocity-specific training adaptations. However, the use of target velocities can also have other implications if using velocity data to estimate 1RM. If a load–velocity regression equation was determined while an athlete was instructed to move as fast as possible (6,9,17,19), it may overestimate an athlete's maximum strength if they are instructed to lift a load with a velocity target. This may happen as the performer may have achieved a faster movement velocity during testing not because of any change in their state, but because of a change in instruction. For example,

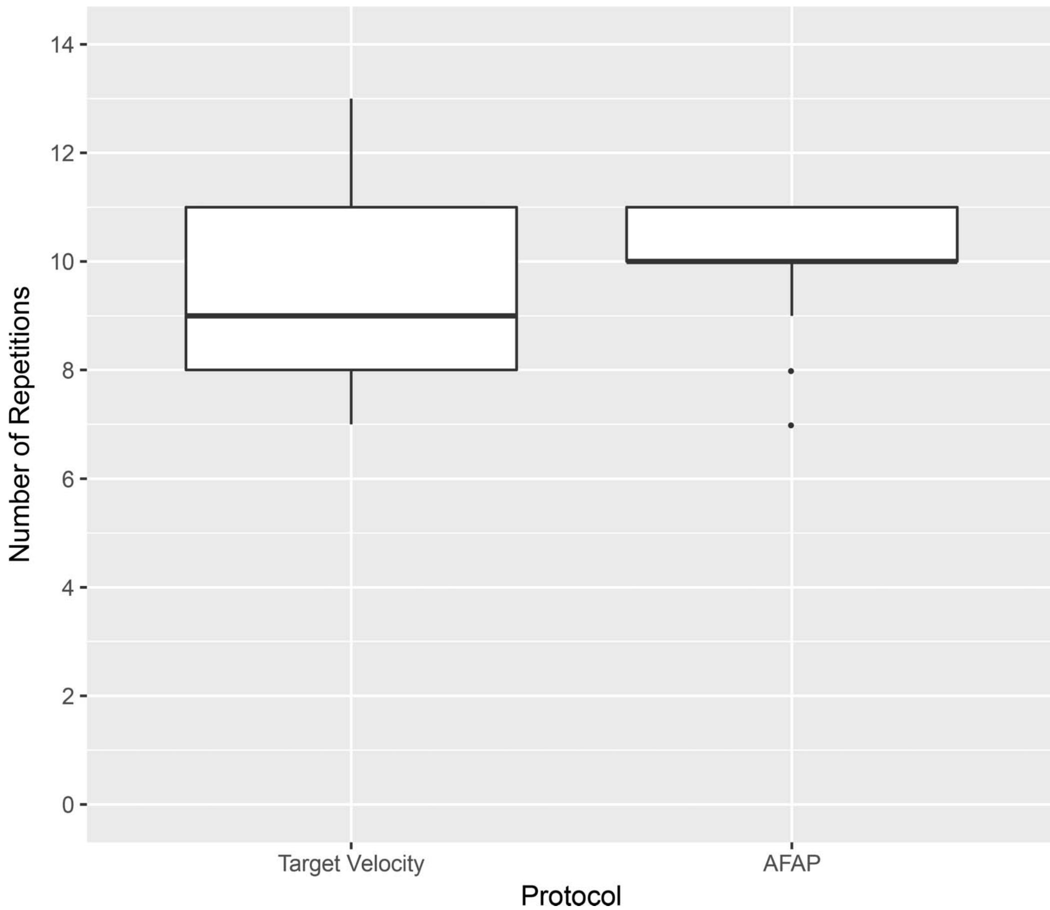

Figure 2. Box plot of participants' repetitions performed during the repetition maximum test with a 75% 1 repetition maximum load after the instruction to attain a target velocity of 1.0 m·s^{-1} and to move as fast as possible (AFAP). There was no statistically significant difference in the number of repetitions performed after the administration of each instruction protocol (target velocity session = 9.61 ± 2.06 repetitions, AFAP session = 10.0 ± 1.29 repetitions, V = 15.5, p = 0.43).

using regression equations for a bench press from Helms et al. (7), a mean velocity of 1.00 m·s^{-1} would correspond to 45% 1RM, but a mean velocity of 0.97 m·s^{-1} would correspond to 47% 1RM. Although this difference is small, it may still be important to provide the same instructions during testing and training if using velocity data to estimate 1RM, depending on how accurate the prediction must be for the exercise professional and their trainee.

There were limitations in this study that should be highlighted to interpret these findings. First, it is important to note the actual movement velocities attained by the participants. Although all participants were lifting 45% of their 1RM, many did not reach the 1.0 m·s^{-1} velocity target. This highlights that with a standardized relative load, a universal velocity target may not be appropriate for a group of athletes, as it may be too challenging for some to attain. Therefore, individualized velocity targets may elicit a more favorable response, and superior high velocity-specific training adaptations. Future research should consider using velocity targets that are specific to individuals' capabilities, and testing athletes that compete at high-velocities as powerlifters compete at relatively lower velocities. In addition, feedback was provided for each repetition, as is done in practice and has been reported in previous research (14,15). However, prior work has also shown that providing KR after each trial may not be as effective when learning a new skill as intermittent KR (1,13). Although intermittent KR may enhance long-term learning, it may come at a detriment to acute performance (1). Therefore, future research should investigate the potential value in instructing athletes with a challenging target velocity and providing intermittent KR. Finally, the average difference in velocity between sessions was 0.02 m·s^{-1}. This small difference may be difficult to detect with some devices, meaning a valid device, such as the GymAware (2), should be used to monitor velocity and provide feedback during training.

Practical Applications

The use of a challenging target velocity was able to elicit faster movement velocities in comparison with instructing participants to move as fast as possible, without reducing the number of repetitions performed in a subsequent RM test. Encouraging participants to attain a challenging target velocity during training may be an effective strategy to use when trying to elicit velocity-specific training adaptations.

Acknowledgments

The authors report no conflict of interest. The results of the present study do not constitute endorsement of the product by the authors or the NSCA.

References

1. Adams JA. A closed-loop theory of motor learning. *J Mot Behav* 3: 111–150, 1971.
2. Banyard HG, Nosaka K, Sato K, Haff GG. Validity of various methods for determining velocity, force, and power in the back squat. *Int J Sports Physiol Perform* 12: 1170–1176, 2017.
3. Coker C. Optimizing external focus of attention instructions: The role of attainability. *J Mot Learn Dev* 4: 116–125, 2016.
4. Erez M, Zidon I. Effect of goal acceptance on the relationship of goal difficulty to perform-mance. *J Appl Psychol* 69: 69, 1984.
5. Frost DM, Cronin JB, Newton RU. A comparison of the kinematics, kinetics and muscle activity between pneumatic and free weight resistance. *Eur J Appl Physiol* 104: 937–956, 2008.
6. González-Badillo JJ, Sánchez-Medina L. Movement velocity as a measure of loading in- tensity in resistance training. *Int J Sports Med* 31: 347–352, 2010.
7. Helms ER, Storey A, Cross MR, et al. RPE and velocity relationships for the back squat, bench press, and deadlift in powerlifters. *J Strength Cond Res* 31: 292–297, 2017.
8. Izquierdo M, González-Badillo JJ, Häkkinen K, et al. Effect of loading on unintentional lifting velocity declines during single sets of repetitions to failure during upper and lower extremity muscle actions. *Int J Sports Med* 27: 718–724, 2006.
9. Jidovtseff B, Harris NK, Crielaard JM, Cronin JB. Using the load-velocity relationship for 1RM prediction. *J Strength Cond Res* 25: 267–270, 2011.
10. Kilduski NC, Rice MS. Qualitative and quantitative knowledge of results: Effects on motor learning. *Am J Occup Ther* 57: 329–336, 2003.
11. Latham GP, Locke EA. Self-regulation through goal setting. *Organ Behav Human Decis Process* 50: 212–247, 1991.
12. Locke EA, Latham GP. New directions in goal-setting theory. *Curr Dir Psychol Sci* 15: 265–268, 2006.
13. Newell KM. Knowledge of results and motor learning. *Jrnl of Motor Behav* 6: 235–244, 1974.
14. Randell AD, Cronin JB, Keogh JW, Gill ND, Pedersen MC. Effect of instantaneous performance feedback during 6 weeks of velocity- based resistance training on sport-specific performance tests. *J Strength Cond Res* 25: 87–93, 2011.
15. Randell AD, Cronin JB, Keogh JW, Gill ND, Pedersen MC. Reliability of performance velocity for jump squats under feedback and non- feedback conditions. *J Strength Cond Res* 25: 3514–3518, 2011.
16. Sale DG. Postactivation potentiation: Role in human performance. *Exerc Sport Sci Rev* 30: 138–143, 2002.
17. Sanchez-Medina L, Perez C, Gonzalez-Badillo J. Importance of the propulsive phase in strength assessment. *Int J Sports Med* 31: 123–129, 2010.
18. Sanchez-Medina L, González-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc* 43: 1725–1734, 2011.
19. Sanchez-Medina L, Pallarés JG, Pérez CE, Morán-Navarro R, González-Badillo JJ. Estimation of relative load from bar velocity in the full back squat exercise. *Sports Med Int Open* 1: E80–E88, 2017.
20. Schmidt RA, Lee TD. 3rd, ed. *Motor Control and Learning: A Behavioral Emphasis*. Champaign, IL: Human Kinetics, 1999. pp. 406–407.
21. Wulf G, Dufek JS. Increased jump height with an external focus due to enhanced lower extremity joint kinetics. *J Mot Behav* 41: 401–409, 2009.
22. Wulf G, Höß M, Prinz W. Instructions for motor learning: Differential effects of internal versus external focus of attention. *J Mot Behav* 30: 169–179, 1998.
23. Wulf G, Dufek JS. Increases in jump-and-reach height through an external focus of attention. *Int J Sports Sci Coach* 2: 275–284, 2007.